24/9/26

Bio Inspired Systems
(BIS, Lab 0)

1> Genetic Algorithm for optimization

Summary:

Genetic Algorithm is a population based optimization technique, inspired by natural selection. It starts with random solutions and improves them using selection, crossover and mutation over several generations

Applications

- Scheduling and timetabling
- Feature selection in Machine Learning
- Engineering design optimization
- Routing and resource allocation (TSP)

Versions

- Simple/Standard GA
- Real Coded GA
- Multi Objective GA

Advantages

- Works well for complex search space
- Doesn't require derivatives
- Can escape local optima

Disadvantages:

- Can be computationally expensive
- Parameter selection
- No guarantee of global optimum

<2> Particle Swarm Optimization

- Summary: PSO is inspired by bird flocking and fish schooling. Particles move through search space using personal best & global best positions

- Applications
  - Function optimization
  - Neural network training

Versions/Variants
- Standard PSO
- Binary PSO

Advantages
- Simple to implement
- Fast convergence

Disadvantages
- Get trapped local optima
- Converge Prematurely

<3> Ant Colony Optimization

Summary: ACO is inspired by foraging behaviour of ants. Artificial ants use pheromone trails to find good path & solutions

Applications
- Travelling Salesman
- Vehicle routing

Add other appln's ?

Versions/Variants:
- Ant Systems (AS)
- Ant Colony System (ACS)

Advantages
- Combinatorial & routing problems effective use

Disadvantages
- Can converge slowly
- get trapped in local optima

<4> Cuckoo Search

Summary: Cuckoo Search is inspired by the brood parasitic behaviour of cuckoo birds it uses Lévy flights to generate new solution and replace poor solution

Applications:

- Engineering Optimization
- Machine Learning

Versions / Variants:

- Standard Cuckoo search
- Adaptive Cuckoo search

Advantages

- Simple algorithm
- Good global search

Disadvantages

- Can be computationally expensive, good global search; converge slowly

<5> Grey wolf Optimizer (GWO)

Summary: GWO is inspired by social hierarchy and hunting behaviour of grey wolves. alpha, beta & delta, wolves lead other wolves to good solutions

Applications:

- Engineering design
- Feature Selection

Variations / Variants

- Standard GWO
- Binary GWO

Advantages

- Simple to implement
- Use few parameters

Disadvantages

- Slow for large problems
- Can converge Prematurely

<6> Parallel Cellular Algorithms (PCA)

Summary: PCA is inspired by Cellular automata and biological cells. The solution space is divided into cells that update are their neighbors & predefined rules

Applications
- Large Scale optimization
- Image Processing

Versions
- Cellular Genetic Algo (CGA)
- Cellular PSO

Advantages
- Suitable for parallel processing
- Suitable for large programs

Disadvantages
- Neighborhood design can be difficult
- Parallel implementation can be complex

<7> Gene Expression Algorithm (GEA)

Summary: GEA is inspired by gene expressions in biological organisms. Solutions are evolved using selection, crossover, mutation & gene expression

Applications
- Engineering optimization
- Machine Learning

Versions / Variants
- Basic gene expression
- gene expression programs (GEP)

Advantages
- Represent complex solution
- Suitable non linear problems

Disadvantages
- Representation can be complex
- Requires Parameter tuning

24/9/25